

Aden Briano

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Education

Rice University – Houston, TX

Bachelor of Science in Electrical & Computer Engineering

GPA: 3.51 / 4.0, Expected Graduation May 2026

Relevant Coursework: Advanced VLSI, High-Performance Computer Architecture, High-Speed System Design, Digital IC Design

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, MATLAB, Assembly

Electrical & Computer Engineering Skills: Verilog, SystemVerilog, KiCad, EAGLE, Arduino, ESP32, MSPM0+, QuestaSim, Magic VLSI, Irsim, Design Compiler, Innovus, LTSpice, HSPICE, Cadence Virtuoso, AMD Vivado, HyperLynx, CUDA, Linux/Unix

Experience

Systems Engineering Intern | General Dynamics Mission Systems – Scottsdale, AZ

May 2025 – Present

- Resolved a CAN communication failure on a \$100K+ digital control module by tracing it to a misrouted LVDS receiver IC, ensuring correct control voltages for frequency conversion modules, and preventing a multi-week unit integration delay
- Authored and redlined test procedures for space-qualified, electrostatic discharge-sensitive digital and RF flight hardware
- Executed functional and performance testing on 14 flight modules, providing integration-ready hardware that relieved schedule pressure by enabling early unit stackup and feature pass-off on flight hardware

Digital Logic Design TA | Rice University ECE Department – Houston, TX

Aug 2025 – Present

- Provided technical guidance to 100+ students on the design and verification of a single-cycle 16-bit processor

Rover Electrical Team Lead | Rice University Robotics Club – Houston, TX

Aug 2024 – Present

- Designed and fabricated a motor controller PCB in KiCad, integrating DRV8825 stepper drivers and an ESP32 to precisely control five stepper motors on a robotic arm
- Designed a power distribution network and communication systems to ensure reliable power delivery and telemetry
- Showcased the Rover design to 600+ students and faculty at the Huff OEDK Engineering Design Showcase

Projects

Snake Game ASIC | Verilog, Magic, QuestaSim, Cadence Innovus, Synopsys Design Compiler

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- Designed a two-phase clock FSM + Datapath architecture in Verilog to emulate the classic Snake game in static CMOS Logic
- Verified modules in QuestaSim, integrated units, synthesized 0.5 μm logic with Design Compiler, & achieved timing closure
- Performed P&R in Cadence Innovus and exported for physical layout integration and final simulation in Magic VLSI

Adder Design Competition | Cadence Virtuoso, HSPICE

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- Designed and benchmarked CMOS Full Adders across logic families, achieving a 54% reduction in propagation delay
- Constructed multiple 32-bit Carry-Square Root Select Adders, reducing delay by 3.3%, lowering average current draw by 58%, and optimizing Energy-Delay Product by 61% through transistor sizing, logical effort analysis, and circuit topology
- Implemented Transmission Gate MUXes and Mirror Full Adders in the fastest CSA design for critical path optimization

Single Cycle 16-Bit Processor | Verilog, FPGA Programming, AMD Vivado, Assembly

github.com/BrianoAden/16-b-Processor

- Developed a single-cycle 16-bit processor in Verilog supporting arithmetic and logic instructions
- Simulated and verified functionality using Vivado waveform viewer, then programmed onto a Real Digital Boolean Board

Dog Treat Launcher | Python, PocketBeagle, Shell Scripting, EAGLE

github.com/BrianoAden/Dog-Treat-Launcher

- Designed and prototyped a game on PocketBeagle in Python, integrating WI-FI and an ML model trained on human speech
- Completed system block diagram, power block diagram, and software framework to outline software & hardware prototyping
- Integrated PocketBeagle and supporting components onto custom PCB designed in EAGLE for increased robustness

Radar Gun | C Programming, MSPS003F3 MCU, KiCad

[Radar Gun](#)

- Designed and iterated on Radar Gun PCB with KiCad, integrating MSPS003F3 MCU and various analog/digital components
- Implemented Finite-State Machine architecture in C to ensure deterministic behavior and measurements for the Radar Gun